1. Determine the equation of the tangent line to $g(x)=\left(\frac{1}{x^3}+1\right)(x-1)$ at $x=-1$.

$g(x) = (x^{-3}+1)(x-1)$

$g'(x) = [-3x^{-4}][x-1] + [1][x^{-3}+1]$

$g'(-1) = (-3)(-2) + (1)(-1+1)$

$= 6$

[redacted] $= \left[\frac{1}{(-1)^3}+1\right]\left[(-1)-1\right]$

$g(-1)$ $= 0$

$(-1,0)$

$m = 6$ [redacted] $6(x+1)$

Equation of tangent: $y - 0 =$ [redacted] $)$

$y =$ [redacted]

$y = 6x+6$

2. If $g(x)=\frac{f(x)}{\sqrt{x-1}}$, where $f(5)=8$, and $f'(5)=-5$, find $g'(5)$.

$$g'(x) = \frac{[f'(x)][\sqrt{x-1}] - \left[\frac{1}{2}(x-1)^{-\frac{1}{2}}\cdot(1)\right][f(x)]}{(x-1)}$$

$$g'(5) = \frac{f'(5)\cdot\sqrt{5-1} - \frac{1}{2}(5-1)^{-\frac{1}{2}}\cdot f(5)}{(5-1)}$$

$$= \frac{(-5)(2) - \frac{1}{2}\left(\frac{1}{2}\right)(8)}{4}$$

$$= \frac{-10-2}{4}$$

$$= -3$$

3. Find the points on the function $f(x)=\frac{x+9}{x+8}$ where the tangent lines pass through the origin.

$f(0) = \frac{9}{8}$ $\therefore$ $(0,0)$ is off the curve

$(0,0)$ and $\left(x, \frac{x+9}{x+8}\right)$

$$M_T = \frac{y_2-y_1}{x_2-x_1} = \frac{\frac{x+9}{x+8}-0}{x-0} = \frac{x+9}{x(x+8)}$$

$$f'(x) = \frac{[1][x+8]-[1][x+9]}{(x+8)^2} = \frac{x+8-x-9}{(x+8)^2} = \frac{-1}{(x+8)^2}$$

$f'(x) = M_T$

$$\frac{-1}{(x+8)^2} = \frac{x+9}{x(x+8)}$$

$-x(x+8) = (x+9)(x+8)^2$

$0 = (x+9)(x+8)^2 + x(x+8)$

$0 = (x+8)[(x+9)(x+8)+x]$

$0 = (x+8)(x^2+\cancel{17x}+72+x)$

$0 = (x+8)(x^2+18x+72)$

$0 = (x+8)(x+12)(x+6)$

$x = \{-8, -12, -6\}$

$f(-8) = \frac{-8+9}{-8+8}$ = undefined

$f(-12) = \frac{-12+9}{-12+8} = \frac{3}{4}$

$f(-6) = \frac{-6+9}{-6+8} = \frac{3}{2}$

@ $(-12, \frac{3}{4})$ and $(-6, \frac{3}{2})$

4. Find the equation of the normal to the curve $f(x)=\sqrt[3]{x^2-1}$ at the point where $x=3$.

$f(x) = (x^2-1)^{\frac{1}{3}}$

$f'(x) = \frac{1}{3}(x^2-1)^{-\frac{2}{3}}(2x)$

$f'(3) = \frac{1}{3}(3^2-1)^{-\frac{2}{3}}[2(3)]$

$= \frac{1}{3}(8)^{-\frac{2}{3}}(6)$

$= \frac{1}{3}(\frac{1}{4})(6)$

$= \frac{1}{2}$

$f(3) = \sqrt[3]{3^2-1}$

$= 2$

Equation of the Normal

$m_\perp = -2 \quad (3,2)$

$y-2 = -2(x-3)$

$y = -2x+6+2$

$y = -2x+8$

5. Let f and g be functions such that $g(x)=\frac{f(x)}{x}$. If $y=2x-3$ is the equation of the tangent to the graph of $f(x)$ at $x=1$, what is the equation of the line tangent to the graph of $g(x)$ at $x=1$?

$g'(x) = [f'(x)][x]^{-1} + [-1][f(x)]$

$g'(1) = f'(1)[1] + (-1)[f(1)]$

$= (2)(1) +$ [redacted] $(-1)\cdot(-1)$

$= 3$

$f'(1) = 2$

$g(1) = \frac{f(1)}{(1)}$

$\therefore g(1) = f(1)$

$y = 2(1)-3$

$y = -1$

$\therefore g(1) = f(1) = -1$

tangent to $g(x)$ @ $x=1$

$m = 3 \quad (1,-1)$

$y+1 = 3(x-1)$

6. Find the points on the curve $f(x)=\frac{x}{x+1}$ where the **normal** line is parallel to $x+y=2$.

$f'(x) = \frac{[1][x+1]-[1][x]}{[x+1]^2}$

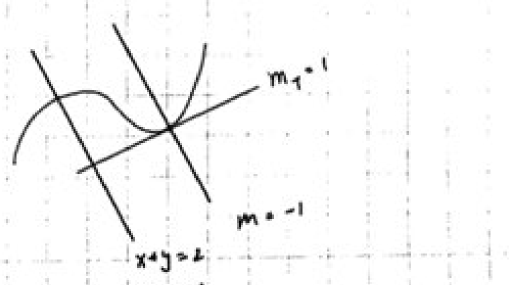
$= \frac{x+1-x}{(x+1)^2}$

$= \frac{1}{(x+1)^2}$

Normal line: $y = -x+2$

$m_\perp = -1$

$m_T = 1$



$\therefore f'(x) = 1$

$\frac{1}{(x+1)^2} = 1$

$1 = (x+1)^2$

$0 = (x+1)^2 - 1$

$0 = [(x+1)+1][(x+1)-1]$

$0 = (x+2)(x)$

$\therefore x = \{0, -2\}$

$f(0) = \frac{0}{0+1}$

$= 0$

$f(-2) = \frac{-2}{-2+1}$

$= \frac{-2}{-1}$

$= 2$

$\therefore$ @ the points $(0,0)$ and $(-2,2)$

7. Let $f(x)=\sqrt{ax^2+b}$. Find values of a and b such that the linear equation $7x+2y=5$ is tangent to $f(x)$ at $x=-1$.

$$f'(x)=\frac{ax}{\sqrt{ax^2+b}}$$

$$f'(-1)=-\frac{7}{2}$$

$$\frac{-a}{\sqrt{a+b}}=\frac{-7}{2} \quad (a \geq 0)$$

$$2a=7\sqrt{a+b} \qquad (1)$$

$$x=-1: \quad 7(-1)+2y=5$$

$$y=6$$

$$f(-1)=6$$

$$\sqrt{a+b}=6 \qquad (2)$$

Sub. (2) into (1):

$$2a=7(6)$$

$$\boxed{a=21} \xrightarrow{\sqrt{a+b}=6} \boxed{b=15}$$

8. Find the area of the triangle determined by the coordinate axes and the tangent to the curve $xy=1$ at $x=1$.

$$y=\frac{1}{x} \quad \text{point}(1,1)$$

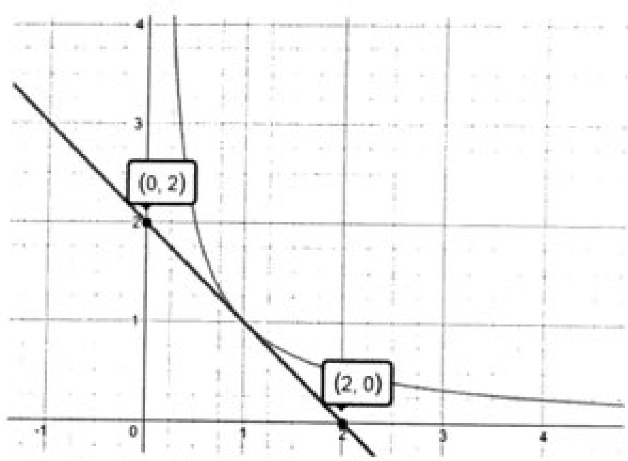
$$y'=-\frac{1}{x^2} \to m_t=-1$$

Equationoftangentline : $y-1=-1(x-1)$

$$y=-x+2$$

x-int$(2,0)$ & y-int$:(0,2)$

Area of tringle : $\frac{1}{2}(2\times 2)=2$ units^2



9 . Consider the curve $y = a\sqrt{x} + \dfrac{b}{\sqrt{x}}$ where a and b are constants. The normal to this curve at the point where $x = 4$ is $4x + y = 22$. Find the values of a and b.

$$y = ax^{\frac{1}{2}} + bx^{\frac{-1}{2}}$$

$$y = x^{\frac{-1}{2}}(ax + b)$$

$$y' = \frac{-1}{2}x^{\frac{-3}{2}}(ax + b) + ax^{\frac{-1}{2}}$$

$$m_T = \frac{-1}{2}(4)^{\frac{-3}{2}}(4a + b) + a(4)^{\frac{-1}{2}}$$

$$m_T = \frac{-1}{16}(4a + b) + \frac{1}{2}a$$

$$m_T = \frac{1}{4}a - \frac{1}{16}b$$

$$4x + y = 22 \quad \uparrow y = -4x + 22$$

$$m_{\perp} = -4 \quad \uparrow m_T = \frac{1}{4}$$

$$\frac{1}{4}a - \frac{1}{16}b = \frac{1}{4} \quad \uparrow 4a - b = 4 \quad (1)$$

$$y = a\sqrt{x} + \frac{b}{\sqrt{x}} \xrightarrow{\text{at } x=4} y = 2a + \frac{b}{2}$$

$$4x + y = 22 \xrightarrow{\text{at } x=4} y = 6$$

$$2a + \frac{b}{2} = 6$$

$$4a + b = 12 \quad (2)$$

$$\left.\begin{aligned} 4a - b &= 4 \quad (1) \\ 4a + b &= 12 \quad (2) \end{aligned}\right\rangle \; a = 2 \,,\; b = 4$$